Notes: Tirole (RES, 1996)

A Theory of Collective Reputations
with Applications to the Persistence of Corruption and to Firm Quality

Contents

1 Big picture

- **Question.** How can a group's reputation be built from individual reputations, and why can bad collective reputations persist even after the original wrongdoers have left?
- **Core mechanism.** Individual behavior is imperfectly observed. Because outsiders cannot perfectly observe an individual's track record, they use the group's past behavior to infer the individual's likely behavior. This creates a two-way feedback: collective reputation affects individual incentives, and individual behavior aggregates into collective reputation.
- **Main idea.** A new member of a group can inherit the "original sin" of earlier members. If the group has behaved badly, outsiders distrust even new members with apparently clean records. Because distrust lowers the payoff from behaving well, new members may choose bad behavior too. This sustains the bad collective reputation.
- **Main applications.** The paper studies two related environments. The first is a *direct exclusion* model motivated by corruption: the trading partner refuses to offer the efficient task when trust is low. The second is a *delegated exclusion* model motivated by firm quality: buyers observe the firm's reputation, while the firm disciplines workers using noisy evidence of individual quality choices.
- **Central results.** The model can generate low-corruption and high-corruption steady states; a one-time corruption shock can push the economy permanently into a high-corruption equilibrium; a short anti-corruption campaign may fail; a sufficiently long campaign or a carefully designed amnesty can restore the good equilibrium; and a firm's product-market competition can weaken workers' incentives to preserve firm quality.
- **Economic takeaway.** Collective reputation is a state variable. It is not just a belief label attached to a group; it changes current incentives and therefore changes future behavior. This is why reputations, stereotypes, corruption, and quality norms can be persistent.

2 Incremental contribution of the paper

- **From individual reputation to collective reputation.** Earlier reputation models often study a single agent. This paper provides a formal model in which a group's reputation is an aggregate of noisy individual reputations.
- **History dependence without common traits.** Stereotypes are usually modeled using either multiple equilibria or an unobserved common group trait. Tirole instead generates stereotypes through history dependence: the behavior of old members affects the incentives of new members.
- **Original sin mechanism.** The paper formalizes the idea that new members may suffer from bad actions committed before they joined. Bad collective history creates distrust, and distrust can make good behavior privately unprofitable.
- **Dynamic complementarity.** The paper identifies a complementarity between past and future behavior. Good past behavior raises trust, which raises the value of behaving well, which supports future good behavior. Bad past behavior does the reverse.
- **Policy contribution.** The paper distinguishes short-lived and sustained interventions. A short anti-corruption campaign may only suppress bad behavior temporarily; a sufficiently long campaign can change beliefs and incentives. Amnesty can be harmful in steady state but welfare improving after a corruption shock.

- **Theory of firm quality.** The paper opens the black box of firm reputation by showing why individual workers may have incentives to maintain the firm's collective reputation when the firm can fire workers based on noisy evidence.
- **Competition insight.** In the firm-quality application, increased product-market competition lowers the rents attached to working in a high-reputation firm. This can make reputation harder to sustain, because workers have less to lose from shirking.

3 Conceptual structure of the paper

3.1 Two variants of collective reputation

Two versions of the collective-reputation mechanism

	Direct exclusion	**Delegated/internal exclusion**
Main application	Corruption and trust between principals and agents	Firm reputation for product quality
Who observes what?	The principal observes an imperfect individual record and the group history	Buyers observe firm quality history; the firm imperfectly observes worker behavior
Who punishes?	The trading partner offers an inefficient task or refuses full trust	The firm fires workers with bad individual evidence
Why group reputation matters	A clean individual record is not fully informative, so group history is used as a signal	Buyers pay according to expected firm quality, so worker rents depend on firm reputation
Central pathology	Corruption can persist because new members inherit distrust	A bad firm reputation can be hard to rebuild without costly mass firing

Interpretation: The two variants have different institutional interpretations, but the mathematical logic is similar. In both cases, imperfect observability of individual behavior is essential. If individual records were perfectly observed, group reputation would add little. If individual records were completely unobserved, individual reputation incentives would collapse.

3.2 Why imperfect observability is central

The model rests on an intermediate-information assumption. The outsider or firm observes enough about an individual's past to make reputation valuable, but not enough to separate all types perfectly. This creates pooling. A member with a clean observed record may be genuinely honest, an opportunist who behaved well, or a bad type whose bad behavior was not detected. Therefore, collective history becomes informative.

Economic message: This is the core reason collective reputation is a social asset. Individual behavior is privately chosen but socially informative. A member who damages his own record also damages the group-level inference environment faced by future members.

4 Direct exclusion model: corruption and trust

4.1 Environment and timing

The economy is stationary and populated by overlapping generations of agents. Agents survive from one period to the next with probability λ. Each exit is replaced by a new entrant, so group size is constant. At each date, each agent meets a new principal and no principal-agent pair meets twice. The principal chooses whether to offer task 1 or task 2.

- **Task 1** is efficient but sensitive to corruption. The principal receives H if the agent is honest and D if the agent cheats.
- **Task 2** is less efficient but safer. The principal receives h if the agent is honest and d if the agent cheats.
- The payoff ordering is
$$H > h \geq d > D.$$
- Task 2 is always weakly worth offering because $d \geq 0$. The relevant question is whether the principal trusts the agent enough to offer task 1.

4.2 Agent types and incentives

There are three types of agents:

- **Honest agents**, with population share a, never cheat.
- **Dishonest agents**, with population share β, always cheat.
- **Opportunists**, with population share γ, trade off current gains from cheating against future reputation losses.

The shares satisfy $a + \beta + \gamma = 1$. An opportunist obtains baseline payoff B from task 1 and b from task 2, with $B > b \geq 0$. Cheating gives a current gain $G > 0$. Let

$$\delta = \lambda \delta_0 \leq 1$$

be the relevant discount factor, combining survival and time discounting.

Interpretation: The opportunist is the strategic type. Honest and dishonest agents create the informational environment: honest agents make a good reputation valuable; dishonest agents prevent clean records from being perfectly revealing.

4.3 Information structure

The principal observes an imperfect, binary track record. If an agent has cheated k times, the principal learns that the agent has cheated at least once with probability χ_k. Otherwise the principal sees a clean record. The key assumptions are

$$\chi_0 = 0 < \chi_1 < \chi_2 < \cdots < 1, \qquad \chi_{k+1} - \chi_k < \chi_k - \chi_{k-1}.$$

The probability of exposure increases with the number of corrupt acts, but at a decreasing rate.

Economic message: The principal does not observe the agent's true age or number of past opportunities to cheat. This is what allows new generations to be pooled with old ones. Without this pooling, new members could separate from the bad past more easily.

5 Steady states in the corruption model

5.1 Low-corruption steady state

A low-corruption steady state is one in which all opportunists behave honestly. Principals offer task 2 to anyone known to have been corrupt, but offer task 1 to agents with clean observed records if the clean record is sufficiently reassuring.

Let

$$Y = (1-\lambda)\left[1 + \lambda(1-\chi_1) + \lambda^2(1-\chi_2) + \cdots\right]$$

be the average probability that a dishonest agent's past corruption remains unobserved. If opportunists always behave honestly, then a clean observed record contains honest agents, opportunists, and dishonest agents whose corruption is undetected. The probability that a clean-record agent will not cheat is

$$\frac{a+\gamma}{a+\gamma+\beta Y}.$$

The principal offers task 1 when

$$\frac{a+\gamma}{a+\gamma+\beta Y}(H-h) + \frac{\beta Y}{a+\gamma+\beta Y}(D-d) > 0.$$

Opportunists must also prefer preserving a clean reputation. If they cheat once and keep cheating, the present discounted chance of being found out is summarized by

$$Z = \chi_1 + \delta\chi_2 + \delta^2\chi_3 + \cdots .$$

The low-corruption steady state requires

$$\frac{G}{1-\delta} \leq \delta(B-b)Z.$$

Economic message: The low-corruption equilibrium is possible when clean records are trusted and opportunists value the future enough to maintain their reputation.

5.2 High-corruption steady state

A high-corruption steady state is one in which opportunists always cheat and principals always offer task 2. In this state, a clean observed record is less reassuring because both opportunists and dishonest agents may have undetected bad histories. The high-corruption steady state exists if

$$\frac{a}{a+(\beta+\gamma)Y}(H-h) + \frac{(\beta+\gamma)Y}{a+(\beta+\gamma)Y}(D-d) < 0.$$

Interpretation: The high-corruption steady state is sustained by distrust. Once task 1 is not offered, the value of a clean record falls, and opportunists have little reason to remain honest.

5.3 Dynamic complementarity

The model can have both low- and high-corruption steady states. The multiplicity is not just a static coordination problem. It comes from a dynamic complementarity:

- good past behavior makes clean records more trusted;
- greater trust raises the payoff from future task 1;

- higher future payoff makes current honesty more attractive;
- current honesty preserves good collective reputation.

Bad behavior reverses the chain.

6 Persistence of corruption and original sin

6.1 One-time shock

The paper studies a simple but powerful experiment. Start in the low-corruption steady state. At date 0, suppose the gain from corruption is temporarily very large, so all opportunists alive at date 0 cheat. From date 1 onward, all fundamentals return to normal. The question is whether the economy returns to low corruption.

Under a simplifying assumption $\chi_1 = \chi_2 = \cdots = \chi$, an opportunist remains corrupt once he has started, and

$$Y = 1 - \lambda\chi, \qquad Z = \frac{\chi}{1-\delta}.$$

Suppose, optimistically, that new opportunists born from date 1 through date t behave honestly. Even then, principals at date t who see a clean record assign probability

$$p(t) = \frac{a + \gamma(1-\lambda^t)}{[a + \gamma(1-\lambda^t)] + [\beta Y + \gamma(1-\chi)\lambda^t]} = \frac{1-\beta-\gamma\lambda^t}{1-\beta\lambda\chi-\gamma\chi\lambda^t}$$

to honest behavior. Let T be the largest date for which this probability is still too low for trust:

$$p(T)(H-h) + (1-p(T))(D-d) < 0.$$

Then $T+1$ is the minimum time needed for suspicion to phase out under the most favorable possible behavior of new entrants.

6.2 Assumption 6 and permanent corruption

If T is large enough, a new opportunist will not find it worthwhile to behave honestly while waiting for trust to return. Assumption 6 states that the gain from cheating during the mistrust period exceeds the discounted value of future trust:

$$G(1+\delta+\cdots+\delta^{T-1}) > \frac{\chi\delta^T(B-b)}{1-\delta} \quad\Longleftrightarrow\quad G(1-\delta^T) > \chi\delta^T(B-b).$$

If this condition holds, the generation born at date 1 cheats. Then the generation born at date 2 also inherits a polluted collective record and cheats. By induction, all later generations cheat.

Economic message: This is the original-sin mechanism. A temporary shock creates enough bad history that even innocent new entrants face low trust. Since good behavior does not pay while trust is absent, new entrants rationally choose bad behavior, reproducing the bad group reputation.

6.3 Anti-corruption campaigns

A short anti-corruption campaign may fail. If the campaign temporarily makes corruption unattractive in one period, opportunists may behave well during the campaign but return to corruption afterward. The campaign does not necessarily restore trust because the group's reputation remains damaged.

A long campaign can work. The minimum campaign length increases with:

- the level of trust required by principals;
- the speed at which suspicion fades;
- the persistence of old generations.

It decreases with:

- the renewal rate of generations;
- the probability of detecting past corruption;
- the value of maintaining a reputation, captured by $(B - b)/G$.

6.4 Amnesty

The paper's treatment of amnesty is subtle. In a low-corruption steady state, amnesty is harmful because it destroys information and weakens incentives. But after an out-of-steady-state corruption shock, an amnesty targeted to the initial corrupt acts can be welfare improving because it removes the stigma that prevents new generations from being trusted.
Economic message: A policy that is harmful in steady state can be helpful after a reputational breakdown. This is because the policy's role changes: in steady state, information disciplines agents; after a shock, old information may mainly perpetuate inefficient distrust.

7 General dynamics and basins of attraction

The paper then relaxes the special assumptions used in the example. For arbitrary histories of corruption, it shows that the game has strategic complementarities: if more opportunists behave honestly, principals are more willing to trust; if principals are more willing to trust, opportunists have stronger incentives to behave honestly.

Under the maintained assumptions, there exists a Pareto-dominant equilibrium. Starting from any history, this equilibrium has one of two forms:

1. it immediately coincides with the high-corruption steady state; or
2. after a finite period during which suspicion is phased out, trust returns and the economy converges to the low-corruption steady state.

Interpretation: The paper does not rely only on multiplicity of equilibria. Even when there are several possible equilibria, it selects the Pareto-dominant one and still obtains path dependence. The relevant state is the distribution of reputational histories.

8 Delegated exclusion: firm reputation for quality

8.1 Set-up

The second application studies a firm as a workers' cooperative. Buyers observe the firm's quality history but do not observe the individual worker who produced a specific item. Workers have individual quality choices, but consumers price the good according to the firm's collective reputation.

Each product has quality H or L, with $H > L > 0$. If consumers believe that the firm produces high quality with probability v_t, the firm's price is

$$p_t = v_t H + (1 - v_t)L.$$

Workers again can be honest, dishonest, or opportunistic. Honest workers always produce high quality; dishonest workers always produce low quality; opportunists compare the effort cost G of producing high quality with the future rent from remaining in the firm.

8.2 High- and low-reputation firms

In a high-reputation firm, opportunists produce high quality and consumers pay

$$p_H = \frac{a+\gamma}{a+\gamma+\beta\widetilde{Y}}H + \frac{\beta\widetilde{Y}}{a+\gamma+\beta\widetilde{Y}}L.$$

The high-reputation equilibrium exists if the rent from remaining in the high-reputation firm is large enough:

$$\frac{G}{1-\delta} \leq \delta p_H \widetilde{Z}.$$

In a low-reputation firm, only honest workers produce high quality. Consumers pay

$$p_L = \frac{a}{a+(\beta+\gamma)\widetilde{Y}}H + \frac{(\beta+\gamma)\widetilde{Y}}{a+(\beta+\gamma)\widetilde{Y}}L < p_H.$$

A low-reputation equilibrium exists if the rent attached to staying in the low-reputation firm is too small to discipline opportunists:

$$\frac{G}{1-\delta} \geq \delta p_L \widetilde{Z}.$$

Economic message: Firm reputation supports worker incentives because workers enjoy rents from belonging to a high-reputation firm. Once the firm has a bad reputation, those rents fall, and low effort becomes harder to deter.

8.3 Hysteresis and mass firing

A firm that loses its reputation may not be able to rebuild it simply through gradual turnover. If opportunists in the existing workforce are locked into low quality and the firm cannot credibly identify all of them, new workers inherit the firm's bad collective reputation. The firm may need mass firing to reset beliefs, but this is costly because it also fires honest workers.

8.4 Competition and firm quality

Let $v > 0$ denote the surplus consumers obtain from an imperfect substitute. Competition effectively lowers the rent from the firm by replacing p_H and p_L with $p_H - v$ and $p_L - v$. The high-reputation condition becomes harder to satisfy:

$$\frac{G}{1-\delta} \leq \delta(p_H - v)\widetilde{Z}.$$

As competition increases, worker rents shrink, so the punishment from losing membership in a high-reputation firm becomes weaker.

Economic message: Competition is usually associated with better discipline, but here it can weaken reputational incentives inside the firm. If workers have less rent to lose, they are less willing to exert costly quality-preserving effort.

9 Relationship to the literature

- **Statistical discrimination and conventions.** The paper differs from static statistical-discrimination models. Those models often rely on multiple equilibria and group-specific equilibrium selection. Tirole instead derives stereotypes from dynamic reputational histories.
- **Common-trait models.** In common-trait models, one member's behavior reveals information about a common group characteristic. In Tirole, there is no fixed common trait; the group reputation is generated by the actual past behavior of members.
- **Repeated-game corporate culture.** Repeated-game models of organizations can support cooperation through punishments. Tirole's model instead emphasizes imperfect individual track records and how current cohorts inherit the reputational state produced by past cohorts.
- **Quality reputation literature.** Classic models of firm quality treat the firm as a unitary actor. Tirole gives microfoundations for firm-level quality by modeling worker incentives and internal exclusion.

10 Key assumptions and results map

Core assumptions and what they do

Object	**Content**	**Role in the model**
Assumption 1	Detection probability rises with past cheating but at a decreasing rate	Generates individual lock-in: once an agent cheats enough, cheating again becomes more attractive
Assumption 2	Clean records are trusted when opportunists behave honestly	Allows a low-corruption steady state
Assumption 3	Future reputation loss outweighs the current gain from cheating	Gives opportunists incentives to stay honest in the low-corruption state
Assumption 4	Clean records are not trusted when opportunists cheat	Allows a high-corruption steady state
Assumption 6	Waiting for trust to return is too costly for new entrants after a corruption shock	Creates permanent corruption after a temporary shock
Assumption 7	A spotty record is always assigned the inefficient task	Simplifies general dynamics and helps establish Pareto-dominant equilibrium
Assumption 3'	High firm reputation creates enough worker rent to induce quality	Supports high-quality firm equilibrium
Assumption 6'	Low firm reputation creates too little rent to induce quality	Supports low-quality firm equilibrium

Main results

Result	Statement	Economic meaning
Proposition 1	The corruption model can have low- and high-corruption steady states	Trust and individual incentives are dynamically complementary
Proposition 2	A one-time corruption shock can permanently push the economy into high corruption; long campaigns or amnesty can restore trust	Bad collective reputation can outlive the original wrongdoers
Proposition 3	From arbitrary histories, the Pareto-dominant equilibrium either remains high corruption or eventually returns to low corruption	The basin of attraction depends on the distribution of reputational histories
Proposition 4	The model extends to internal exclusion by the group; competition may weaken firm quality	Firm reputation is hard to rebuild and depends on worker rents

11 Extracted equations and visual panels from the paper

This is a theory paper and contains no empirical tables or conventional plots. Therefore, the extracted visuals below reproduce the original title/abstract, model primitives, key assumptions, equations, propositions, and firm-quality application panels directly from the paper.

11.1 Title, abstract, and conceptual building blocks

Read: The first panel gives the title, abstract, and the paper's central thesis: collective reputation is modeled as an aggregate of individual reputations. The second panel lists the building blocks: group reputation depends on member behavior; individual behavior is imperfectly observed; group history predicts member behavior; and new cohorts can inherit old reputations.
Economic message: The theory starts from a simple informational externality. Individual behavior is privately chosen but partly pooled through collective beliefs.

A Theory of Collective Reputations (with applications to the persistence of corruption and to firm quality)

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First version received June 1993; *final version accepted August* 1995 (*Eds.*)

The paper is a first attempt at modelling the idea of group reputation as an aggregate of individual reputations. A member's current incentives are affected by his past behaviour and, because his track record is observed only with noise, by the group's past behaviour as well. The paper thus studies the joint dynamics of individual and collective reputations and derives the existence of stereotypes from history dependence rather than from a multiplicity of equilibria or from the existence of a common trait as is usually done in the literature. It shows that new members of an organization may suffer from an original sin of their elders long after the latter are gone, and it derives necessary and sufficient conditions under which group reputations can be rebuilt. Last, the paper applies the theory to analyse when a large firm can maintain a reputation for quality.

1. COLLECTIVE REPUTATIONS

Collective reputations play an important role in economics and the social sciences. Coun-

(a) Title and abstract

(b) By contrast with group belonging, *individual past behaviour is imperfectly observed.* If past individual behaviour was fully unobserved, members of the group would have no incentive to sustain their own reputation and therefore the group would always be expected to behave badly. Conversely, the collective reputation would play no role if individual behaviours were perfectly observed. Imperfect observability of individual behaviour thus underlies the phenomenon of collective reputation.

(c) *The past behaviour of the member's group conditions the group's current behaviour and therefore can be used to predict the member's individual behaviour*. Each member's welfare and incentives are thus affected by the group's reputation.

(d) If we further assume that the age in the group, or the frequency of interactions with the group, or the number of past opportunities for cheating are imperfectly observed, cohorts in a self-regenerating group are partly pooled, and therefore, *the behaviour of new members of a group depends on the past behaviour of their elders*.

We offer two variants of the same model. In the first, a member's individual reputation is imperfectly observed by his potential "trading partner" (who may or may not belong to the group). The incentive to sustain an individual reputation stems from the member's fear of *direct exclusion* by the trading partner. By this we mean that the individual reputation may induce the trading partner to behave in a way undesirable for the member (e.g. by not trading), while the belonging to the group is not affected. We apply the direct exclusion variant to the issue of corruption[1] to explain why corruption is a societal phenomenon and why it tends to persist.

This direct exclusion variant does not seem appropriate when the trading partner has a low probability of knowing the member's past behaviour. The buyer of a car does not even know the names of the worker, foreman and engineer who built the car. Yet brand image is an important factor in the car market. The reason why the car manufacturer's employee has an incentive to maintain quality is the fear of *delegated or internal exclusion*: It may be in the interest of the firm to fire employees who have demonstrated undesirable traits.

(b) Collective-reputation building blocks

11.2 Corruption model primitives and information structure

Read: These panels show the direct-exclusion model. The principal chooses between an efficient trust-sensitive task and a safer inefficient task. Agents are honest, dishonest, or opportunistic. The information structure states that the principal observes only a noisy binary signal of past corruption.

Economic message: The direct-exclusion model turns trust into an allocation rule. Reputation determines whether an agent receives the efficient task.

3. INDIVIDUAL AND SOCIAL STIGMAS: THE CASE OF TRUST

3.1. *The model*

This section develops a simple model in which the efficient organization of economic activity requires a minimum level of trust between contracting parties. More precisely, a principal (the buyer of a service) will contract with an agent (the supplier of the service) only if she is sufficiently confident that the agent will not engage in corrupt activities. The principal has some, albeit imperfect, information about the agent's track record, namely about whether the agent has engaged in corrupt activities in the past.

Matching. We consider a large stationary economy ($-\infty < t < \infty$) in which agents alive at date t remain in the economy up to (at least) date $t+1$ with probability $\lambda \in (0, 1)$. With this "Poisson death process", we assume that each quit is offset by the arrival of a new agent, so that the population of agents is constant. The model is a matching model. Agents have no chance to meet the same principal twice. At each date t, each (alive) agent is matched with a new principal[7]. The principal decides whether to offer task 1 or task 2 to the agent. Task 1 is the efficient task. Task 2 is a less efficient task, but, for the principal, it is less sensitive to the agent's choosing to be corrupt. (In a slightly different version of the model, task 2 corresponds to the absence of a hire.) We will make an assumption guaranteeing that it is always optimal for the principal to at least offer task 2 to the agent rather than not hiring him. Once hired, the agent chooses whether to engage in the corrupt activity, that is whether "to cheat" (behave dishonestly). The principal's payoff from task 1 in the period is H if the agent behaves honestly and D if he cheats. Similarly her payoffs from task 2 are h and d. That task 1 is more sensitive to corruption than task 2 (given that the principal faces a non-trivial choice) means that

$$H > h \geqq d > D.$$

We also assume that $d \geqq 0$ so that it is optimal to hire the agent.

Agents' preferences. There are three types of agents: "honest", in proportion α, "dishonest", in proportion β, and "opportunistic", in proportion γ, where $\alpha + \beta + \gamma = 1$. The proportions are the same for each cohort and therefore for the entire population. Honest agents have a strong distaste for and never engage in corrupt activities (alternatively, if corruption has a probability of being exposed and directly punished, "honest" agents might be ones for whom being punished is very costly). Dishonest agents always cheat, for instance because they derive a high benefit from it (alternatively, in a slightly different model, they might be transient agents who do not care about their reputation). Because honest and dishonest agents behave mechanistically (never and always cheat, respectively), the focus of our analysis is on opportunists.[8] These have no aversion to being corrupt,

(a) Model primitives: tasks, payoffs, and types

model is that there is no hard evidence that could lead to the indictment of a corrupt agent. Alternatively, G could be an expected gain from being corrupt, which would allow a probability of confronting legal sanctions. Last, the agents' discount factor is δ_0. We will let $\delta \equiv \delta_0 \lambda \leqq 1$ denote the "relevant discount factor".

Information. Agents know their own preferences (that is, their types). Principals know the proportions α, β, γ and imperfectly observe the track record of the agent they are matched with. There are several ways of formalizing the imperfect observability of the track record. We choose a simple one in order to illustrate easily the main ideas. The principal has probability x_k of finding out that the agent has engaged in the past at least once in a corrupt activity when the agent has in fact cheated k times[10]. So the *observed* track record, that is the information of the principal the agent is matched with, is binary. The principal knows that the agent has been corrupt at least once, or has no such knowledge.[11] The assumption that the principal does not know the agent's age is important for the effect unveiled in Section 3.3 and giving rise to everlasting effects of a one-time shock in corruption. Of course this assumption should not be taken too literally. It is a metaphor for the idea that the principal may not be fully informed about the number of times the agent had an opportunity to be corrupt in the past or about the length of the member's relationship with the group.

Assumption 1.

$$x_0 = 0 < x_1 < x_2 < x_3 < \cdots < 1 \quad \text{and} \quad x_{k+1} - x_k < x_k - x_{k-1} \quad \text{for all } k.$$

Assumption 1 says that the leakage of information about corruption becomes more likely when the agent has cheated more in the past; and that this increase occurs at a decreasing rate. The second part of this assumption simplifies the analysis by guaranteeing that an individual is locked into corruption after having been corrupt a certain number of times.

(b) Noisy track records and Assumption 1

11.3 Low- and high-corruption steady states

Read: The low-corruption panel defines the clean-record probability and the incentive condition for opportunists to remain honest. The high-corruption panel shows that when opportunists are expected to cheat, clean records are less informative and principals may always choose the inefficient task.

Economic message: The same society can support either good or bad collective behavior depending on the reputational state. The steady states differ not because fundamentals differ, but because the informational environment and incentives differ.

will be the focus of this section, and one is in mixed strategies.[12] It is worth emphasizing that the possible multiplicity of steady states, while interesting in its own right, is not the primary focus of the paper, and that, for given initial conditions, there may be a unique equilibrium even when there are several steady states, as we will see. Furthermore, we will later select the Pareto-dominant equilibrium when there are multiple equilibria.

(*a*) *Low-corruption steady state.* Suppose that all opportunists always behave honestly. A principal offers task 2 to an agent who she knows has been corrupt in the past, since the agent is necessarily a dishonest agent and since $d>D$. In contrast, when the principal has no such information, the agent may be honest or opportunistic, or else be a dishonest agent with a deceivingly clean observed track record. The proportion of honest and opportunistic agents in the economy is $(\alpha+\gamma)$. The proportion of dishonest agents with a clean track record is βY where Y is the average probability that past corruption activities go unnoticed[13]:

$$Y=(1-\lambda)[1+\lambda(1-x_1)+\lambda^2(1-x_2)+\cdots+\lambda^k(1-x_k)+\cdots].$$

The probability that the agent will not cheat given a clean observed record is $(\alpha+\gamma)/(\alpha+\gamma+\beta Y)$. The principal offers task 1 if and only if the following assumption holds:[14]

Assumption 2.

$$\frac{\alpha+\gamma}{\alpha+\gamma+\beta Y}(H-h)+\frac{\beta Y}{\alpha+\gamma+\beta Y}(D-d)>0.$$

Do opportunists have an incentive not to become corrupt? By never being corrupt, they keep a clean (real and observed) record and are always offered task 1. Their payoff is therefore $B+\delta B+\delta^2 B+\cdots=B/(1-\delta)$. Suppose that they instead cheat today and keep cheating in the future. Their expected payoff is then

$$(B+G)+\delta(B+G)\left[\frac{1}{1-\delta}-Z\right]+\delta(b+G)Z,$$

where

$$Z=x_1+\delta x_2+\delta^2 x_3+\cdots$$

is the present discounted probability of being found out in the future given that one has cheated once and will continue cheating. So, a necessary condition for a low-corruption

12. The reader will check that when the two pure-strategy equilibria co-exist, the mixed-strategy equilibrium has the following characteristics. When faced with an agent with a spotless record, principals offer task 2 with probability θ and task 1 with probability $1-\theta$. An agent with a spotty record is offered task 2. The parameter θ satisfies

(a) Low-corruption steady state

$$\frac{G}{1-\delta}\leqq\delta(B-b)Z.$$

Appendix 1 shows that *the low-corruption steady state indeed exists under Assumptions* 1 *through* 3. The intuition is that from Assumption 1, the agent has more incentive to be corrupt, the more he has been corrupt in the past. In this sense, *agents are locked into corruption once they start being corrupt.*

Note also that a low-corruption steady state exists only if the principals are not poorly informed[15]. Agents must have enough incentives to maintain their reputation for honesty.

(*b*) *High-corruption steady state.* Suppose now that opportunists are always corrupt and principals always offer task 2. Because keeping a clean slate has no value, it is indeed optimal for opportunists to be always corrupt. Is is optimal for a principal to offer task 2 to an agent with a clean slate? Such an agent is honest with probabliity $\alpha/[\alpha+(\beta+\gamma)Y]$ and either opportunistic or dishonest with probability $(\beta+\gamma)Y/[\alpha+(\beta+\gamma)Y]$. We thus make

Assumption 4.

$$\frac{\alpha}{\alpha+(\beta+\gamma)Y}(H-h)+\frac{(\beta+\gamma)Y}{\alpha+(\beta+\gamma)Y}(D-d)<0.$$

The high-corruption steady state exists if and only if Assumption 4 holds. Note that Assumption 4 holds when there are enough opportunistic and dishonest agents and when the principals' information is not very precise. The role of imperfect observability is highlighted by the facts that Assumption 3 is violated if the principals' information is very bad and that Assumption 4 is violated if the principals' information is very good and λ is close to 1. The reader can also check that an increase in the population's renewal rate $1/\lambda$ makes it harder for Assumptions 2 and 3 to be satisfied (and so for the low-corruption steady state to exist) and easier for Assumption 4 to hold (and so for the high-corruption steady state to exist).

Proposition 1. (*Steady states*). *The economy has three steady states when Assumptions 2, 3 and 4 are satisfied (and one otherwise). The multiplicity of steady states stems from the dynamic complementarity between past and future reputations. Due to the (endogenous) hysteresis in individual behaviours, good collective behaviour in the past increases trust in the future and thus raises individual incentives to maintain a reputation, resulting in collective reputation being maintained at a high level.*

(b) High-corruption steady state and Proposition 1

11.4 Persistence, campaigns, and amnesty

Read: The first panel gives the key persistence experiment: a temporary shock corrupts date-0 opportunists, after which clean records are no longer trusted for a number of periods. Assumption 6 says new entrants prefer cheating rather than waiting for trust to return. The second panel shows why a one-period anti-corruption campaign may fail and how the minimum campaign length depends on model parameters.

Economic message: Bad history can create a self-perpetuating reputational trap. Restoring good behavior requires not only stopping current corruption but also rebuilding trust long enough for honest incentives to become privately valuable again.

That is, the probability of exposure to corrupt activities is independent of the number of past corrupt acts. Assumption 5, which is a limit case of Assumption 1, implies in particular that an opportunist remains corrupt once he has started; it also implies that $Y=1-\lambda x$ and $Z=x/(1-\delta)$.

We make Assumptions 2 through 4. Let the economy start in the low-corruption steady state (see Section 3.4 for a justification). Suppose now that the economy faces a temporary shock at date 0. The gain from being corrupt at that date is very large, and so all opportunistic agents alive at date 0 get corrupt. The parameters of the model (including the gain G from cheating) are unchanged at dates $1, 2, \ldots$. We show that under an additional assumption, the economy cannot go back to the low-corruption steady state. Indeed, the *unique* continuation equilibrium exhibits a high level of corruption forever.

Let us perform the following thought experiment. Suppose that the opportunistic agents born at date 1 through t behave honestly before and at date t. This presumption gives the best chance to the existence of trust at date t. The probability of honest behaviour at date t given an observed clean record and given that opportunists born at or before date 0 are locked into corruption is

$$p(t)\equiv\frac{\alpha+\gamma(1-\lambda)(1+\lambda+\cdots+\lambda^{t-1})}{[\alpha+\gamma(1-\lambda)(1+\lambda+\cdots+\lambda^{t-1})]+[\beta Y+\gamma(1-x)(1-\lambda)(\lambda^t+\lambda^{t+1}+\cdots)]}$$
$$=\frac{\alpha+\gamma(1-\lambda^t)}{[\alpha+\gamma(1-\lambda^t)]+[\beta Y+\gamma(1-x)\lambda^t]}=\frac{1-\beta-\gamma\lambda^t}{1-\beta\lambda x-\gamma x\lambda^t}.$$

Recall that $p(1)(H-h)+(1-p(1))(D-d)<0$ (this is Assumption 4) and that $p(\infty)(H-h)+(1-p(\infty))(D-d)>0$ (this is Assumption 2). Noting that p is an increasing function, we let T denote the largest t such that

$$p(T)(H-h)+(1-p(T))(D-d)<0. \qquad (1)$$

That is, under the most optimistic assumption, principals still do not trust agents with observed clean records at date T; thus $(T+1)$ is the minimum length of time for suspicion to be phased out. Suppose now that

Assumption 6.

$$G(1+\delta+\cdots+\delta^{T-1})>\frac{x\delta^T(B-b)}{1-\delta}\Leftrightarrow G(1-\delta^T)>x\delta^T(B-b).$$

Assumption 6 states that it is a dominant strategy for an agent born at date 1 to cheat at date 1 (and therefore forever) given that the agent will not be trusted before (at best) date $(T+1)$. The left-hand side of Assumption 6 is the gain from cheating from date 1 through date T (discounted at date 1), and the right-hand side is an upper bound on the cost of not being offered task 1 after date $(T+1)$. Note that Assumption 6 requires T not to be too small, since with x_k constant for $k\geqq 1$, Assumption 3 is equivalent to $G\leqq x\delta(B-b)$.

Consider now the generation born at date 2. All its elders have been corrupt in the past, and Assumption 6 ensures similarly that cheating at date 2 and thereafter is a dominant strategy. By induction, the same is true for all generations. *Corruption has*

(a) Original sin and Assumption 6

economy can return to the low-corruption steady state in the long-run: Just specify that all opportunistic agents joining the group after date 0 behave honestly and that principals offer task 2 from date 1 through date T, and offer task 1 (for a spotless record) from date $T+1$ on. This equilibrium converges to the low-corruption steady state as $t\to\infty$.

This simple model also illustrates the possible failure of a short-run anti-corruption campaign. Suppose that at date 1 (or, equivalently at any later date) the government runs a tough anti-corruption campaign that lasts one period and makes it unprofitable for opportunists to engage in corruption at that date. Suppose further that the following strengthening of Assumption 6 holds:

$$G(1+\cdots+\delta^{T-2})\geqq x\delta^{T-1}(B-b)/(1-\delta).$$

Then it is a dominant strategy for generations born at dates 1 and 2 to cheat at date 2, and corruption prevails at all dates after date 1. *The anti-corruption campaign only implies a decrease in corruption during the campaign and has no effect thereafter. Corruption does not ratchet down.*

Comparative statics. Condition (1) defines the minimum length of time for suspicion to phase out. Let $T=T(\lambda, x, \rho)$ where $\rho\equiv(d-D)/[(H-D)-(h-d)]$ is the smallest probability of honest behaviour that makes a principal choose task 1. T is increasing in λ and ρ and decreasing in x: The suspicion takes longer to phase out, the slower the rate of renewal of generations, the higher the level of trust required by principals, and the smaller the probability of detection of past corrupt behaviour.

Thus, perpetual corruption (that is, an equilibrium coinciding with the high-corruption steady state from date 1 on) is the unique continuation equilibrium if (rewriting Assumption 6):

$$T(\lambda, x, \rho)>\frac{\log\left(1+\dfrac{x(B-b)}{G}\right)}{\log\left(\dfrac{1}{\delta}\right)}.$$

We see that the increase in the probability x of detection not only lowers T, but also raises the young generations' incentive to be honest, so both effects reinforce each other. As for the effect of λ, we must keep $\delta=\lambda\delta_0$ constant to obtain an unambiguous conclusion; for, if we keep the interest rate (δ_0) constant, an increase in λ has two opposite effects: It makes young generations more eager to maintain a good reputation, but it also slows down the renewal of generations.

We can alternatively state our results in terms of the smallest number τ of periods of anti-corruption campaigns needed to allow a (long-run) return to the low-corruption steady state. (An anti-corruption campaign at date t is defined as one in which the probability of being caught and the resulting penalties are high enough so that being corrupt at date t is a dominated strategy for an opportunist. Dishonest agents still cheat during the campaign.) A prolonged anti-corruption campaign from date 1 through date τ allows a (long-run) return to the low-corruption steady state if and only if opportunists with a spotless record at $\tau+1$ find it optimal to be honest conditionally on trust being restored

(b) Campaign failure and comparative statics

11.5 Propositions on dynamics

Read: Proposition 2 summarizes the main persistence result: without a sufficiently strong intervention, the high-corruption steady state is the unique continuation equilibrium after the shock; long campaigns or amnesty can restore the good state. Proposition 3 extends the analysis to arbitrary histories and characterizes the Pareto-dominant equilibrium.

Economic message: The paper's dynamic result is stronger than a mere multiple-equilibrium story. Starting from a given history, reputational dynamics can determine whether the economy is trapped or recovers.

is the smallest integer τ such that:

$$T(\lambda, x, \rho) - \tau \leq \frac{\log\left(1 + \frac{x(B-b)}{G}\right)}{\log\left(\frac{1}{\delta}\right)}.$$

Amnesty. Consider now a policy of amnesty for past corrupt behaviour (if feasible). One may for example have in mind a strict libel law that severely punishes those who communicate to a principal evidence about an agent's past corrupt behaviour (alternatively, an amnesty could exempt the corrupt agent from a jail sentence in the variant in the spirit of Section 4, in which punishments are inflicted by the group itself.)

Amnesty is irrelevant in a high-corruption steady state, in which there is no trust anyway. And it can only be detrimental if the low-corruption steady state prevails, because it destroys the information principals use to give task 2 to some of the corrupt agents, and (for a sustained amnesty) because it destroys opportunistic agents' incentives to behave honestly.

By contrast, an amnesty may be welfare enhancing out of steady state. An amnesty limited to corrupt acts performed at date 0 enables the group to return immediately to the low-corruption steady state. (Of course, the issue is whether such an amnesty is feasible. It certainly is possible to exempt corrupt acts at date 0 from jail sentences or fines, but it may be harder for a government to dispel mistrust among citizens through a governmental measure such as a libel law.)

We summarize the results of this section in:

Proposition 2. *Consider a one-time shock in corruption (all opportunistic agents are corrupt at date 0). Under Assumptions 2 through 6:*

(a) in the absence of anti-corruption campaigns or amnesty, the high-corruption steady state (perpetual corruption) is the unique continuation equilibrium;

(b) the minimum number of periods of anti-corruption campaigns (discouraging opportunists from being corrupt during the period) required to allow a (long-run) return to the low-corruption steady state increases with the level of trust ρ required by principals, and decreases with the rate of renewal of generations (keeping the agents' discount factor constant), with the probability of detection of corruption, and with the agents' "eagerness" $(B-b)/G$ to maintain a reputation;

(c) an amnesty limited to corrupt acts performed at date 0 enables the group to return immediately to the low-corruption steady state, and yields a Pareto improvement relative to

(a) Proposition 2 and amnesty

tunistic agent who has cheated k times before date t behaves honestly at date t.[18] To simplify the analysis, we make an assumption that guarantees that it is a dominant strategy for a principal to offer task 2 when confronted with an agent with a spotty record:

Assumption 7.

$$\frac{\gamma}{\beta+\gamma}(H-h) + \frac{\beta}{\beta+\gamma}(D-d) < 0.$$

(Note that for any history, $\beta/(\beta+\gamma)$ is a lower bound on the probability that the agent with a spotty record is dishonest and therefore on the probability that he will be corrupt.)

The game exhibits strong strategic complementarities. *Heuristically*, for all t, a weak increase in θ_{kt} for some k weakly increases the reaction correspondence μ_t, and has no impact on the other reactions $\{\mu_\tau\}_{\tau<t}$ (and has no "cross-effects" on $\{\theta_{k't'}\}_{(k't' \neq kt)}$). Conversely, an increase in μ_t weakly increases the reaction correspondences $\theta_{kt'}$ for all k and $t' < t$, and has no impact on others. Because of these strategic complementarities, it turns out that one equilibrium is preferred by the principals and by all agents (regardless of their types and histories). (We have not quite demonstrated that the game is supermodular. We have not shown that an increase in θ_{kt} for some k weakly increases the reactions $\{\mu_\tau\}_{\tau>t}$. A sufficient condition for this would be that $\theta_{\cdot t'}$ be weakly decreasing for all $t' \in \{t+1, \ldots, \tau\}$. We have not proved this generalization of the steady-state property demonstrated in Appendix 1. Yet we conjecture that the game can be made supermodular. If this conjecture turns out to be correct then part (a) of Proposition 3—the existence of a Pareto-dominant equilibrium—is simply Topkis' theorem (1979).)

When there are multiple equilibria (which need not be the case: see Section 3.3), we thus select, we feel in a reasonable way, a unique equilibrium. This illustrates again the sharp contrast in approach with the theory of conventions (including the statistical theory of discrimination) which makes the multiplicity of equilibria the focus of the analysis and assigns different equilibria to different groups. We now ask whether the equilibrium converges to one of the three steady states and we characterize its dynamics:

Proposition 3. *Under Assumptions 1, 2, 3, 4 and 7, for any history of corruption before date 1, starting at date 1:*

(a) there exists a Pareto-dominant equilibrium,

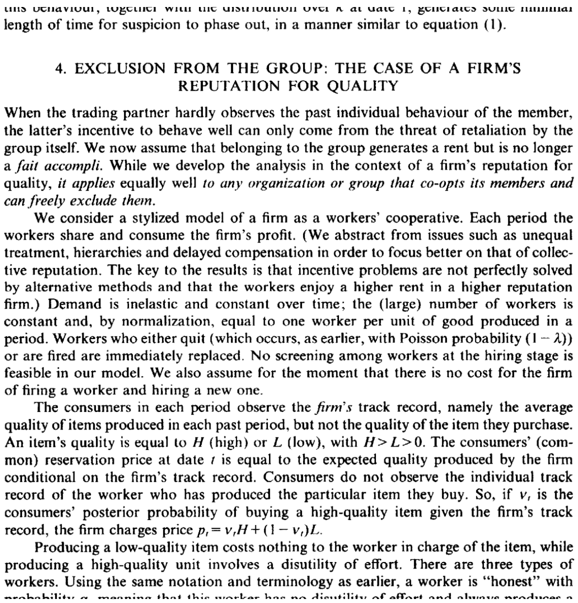
(b) either the equilibrium is unique and coincides from date 1 on with the high-corruption

(b) Proposition 3: general basins of attraction

11.6 Firm quality application

Read: These panels show the delegated-exclusion model. Buyers observe the firm's track record, not the individual worker's record. The firm may fire workers when it detects low-quality behavior. High- and low-reputation firm equilibria depend on whether firm rents are sufficient to induce opportunistic workers to produce high quality.

Economic message: The model provides microfoundations for brand reputation. A brand is valuable because it creates rents that discipline workers; if the brand is damaged, those rents fall and low quality becomes harder to deter.



length of time for suspicion to phase out, in a manner similar to equation (1).

4. EXCLUSION FROM THE GROUP: THE CASE OF A FIRM'S REPUTATION FOR QUALITY

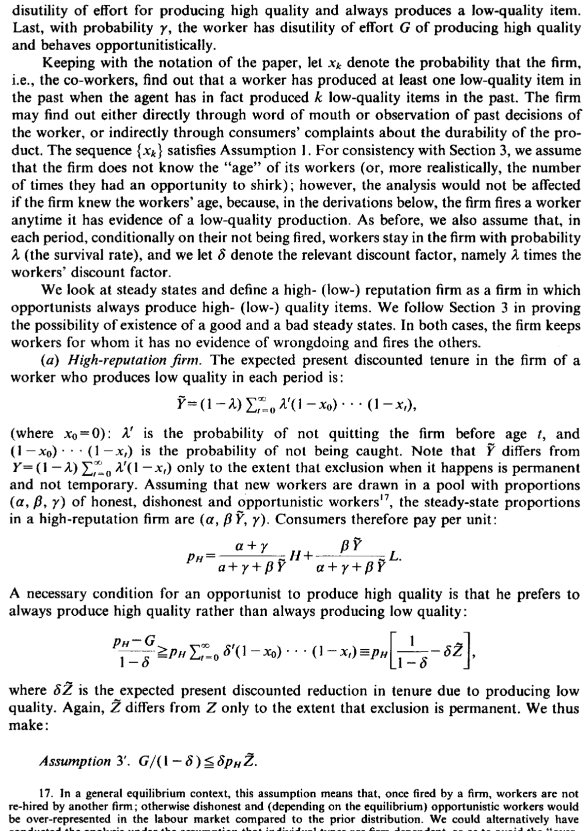
When the trading partner hardly observes the past individual behaviour of the member, the latter's incentive to behave well can only come from the threat of retaliation by the group itself. We now assume that belonging to the group generates a rent but is no longer a *fait accompli*. While we develop the analysis in the context of a firm's reputation for quality, *it applies* equally well *to any organization or group that co-opts its members and can freely exclude them*.

We consider a stylized model of a firm as a workers' cooperative. Each period the workers share and consume the firm's profit. (We abstract from issues such as unequal treatment, hierarchies and delayed compensation in order to focus better on that of collective reputation. The key to the results is that incentive problems are not perfectly solved by alternative methods and that the workers enjoy a higher rent in a higher reputation firm.) Demand is inelastic and constant over time; the (large) number of workers is constant and, by normalization, equal to one worker per unit of good produced in a period. Workers who either quit (which occurs, as earlier, with Poisson probability $(1-\lambda)$) or are fired are immediately replaced. No screening among workers at the hiring stage is feasible in our model. We also assume for the moment that there is no cost for the firm of firing a worker and hiring a new one.

The consumers in each period observe the *firm's* track record, namely the average quality of items produced in each past period, but not the quality of the item they purchase. An item's quality is equal to H (high) or L (low), with $H>L>0$. The consumers' (common) reservation price at date t is equal to the expected quality produced by the firm conditional on the firm's track record. Consumers do not observe the individual track record of the worker who has produced the particular item they buy. So, if v_t is the consumers' posterior probability of buying a high-quality item given the firm's track record, the firm charges price $p_t = v_t H + (1-v_t)L$.

Producing a low-quality item costs nothing to the worker in charge of the item, while producing a high-quality unit involves a disutility of effort. There are three types of workers. Using the same notation and terminology as earlier, a worker is "honest" with

(a) Firm quality setup



disutility of effort for producing high quality and always produces a low-quality item. Last, with probability γ, the worker has disutility of effort G of producing high quality and behaves opportunitistically.

Keeping with the notation of the paper, let x_k denote the probability that the firm, i.e., the co-workers, find out that a worker has produced at least one low-quality item in the past when the agent has in fact produced k low-quality items in the past. The firm may find out either directly through word of mouth or observation of past decisions of the worker, or indirectly through consumers' complaints about the durability of the product. The sequence $\{x_k\}$ satisfies Assumption 1. For consistency with Section 3, we assume that the firm does not know the "age" of its workers (or, more realistically, the number of times they had an opportunity to shirk); however, the analysis would not be affected if the firm knew the workers' age, because, in the derivations below, the firm fires a worker anytime it has evidence of a low-quality production. As before, we also assume that, in each period, conditionally on their not being fired, workers stay in the firm with probability λ (the survival rate), and we let δ denote the relevant discount factor, namely λ times the workers' discount factor.

We look at steady states and define a high- (low-) reputation firm as a firm in which opportunists always produce high- (low-) quality items. We follow Section 3 in proving the possibility of existence of a good and a bad steady states. In both cases, the firm keeps workers for whom it has no evidence of wrongdoing and fires the others.

(*a*) *High-reputation firm*. The expected present discounted tenure in the firm of a worker who produces low quality in each period is:

$$\tilde{Y} = (1-\lambda)\sum_{t=0}^{\infty} \lambda^t (1-x_0)\cdots(1-x_t),$$

(where $x_0 = 0$): λ^t is the probability of not quitting the firm before age t, and $(1-x_0)\cdots(1-x_t)$ is the probability of not being caught. Note that $\tilde{Y}$ differs from $Y = (1-\lambda)\sum_{t=0}^{\infty}\lambda^t(1-x_t)$ only to the extent that exclusion when it happens is permanent and not temporary. Assuming that new workers are drawn in a pool with proportions (α, β, γ) of honest, dishonest and opportunistic workers[17], the steady-state proportions in a high-reputation firm are $(\alpha, \beta\tilde{Y}, \gamma)$. Consumers therefore pay per unit:

$$p_H = \frac{\alpha+\gamma}{\alpha+\gamma+\beta\tilde{Y}} H + \frac{\beta\tilde{Y}}{\alpha+\gamma+\beta\tilde{Y}} L.$$

A necessary condition for an opportunist to produce high quality is that he prefers to always produce high quality rather than always producing low quality:

$$\frac{p_H - G}{1-\delta} \geqq p_H \sum_{t=0}^{\infty} \delta^t (1-x_0)\cdots(1-x_t) \equiv p_H\left[\frac{1}{1-\delta} - \delta\tilde{Z}\right],$$

where $\delta\tilde{Z}$ is the expected present discounted reduction in tenure due to producing low quality. Again, $\tilde{Z}$ differs from Z only to the extent that exclusion is permanent. We thus make:

Assumption 3'. $G/(1-\delta) \leqq \delta p_H \tilde{Z}$.

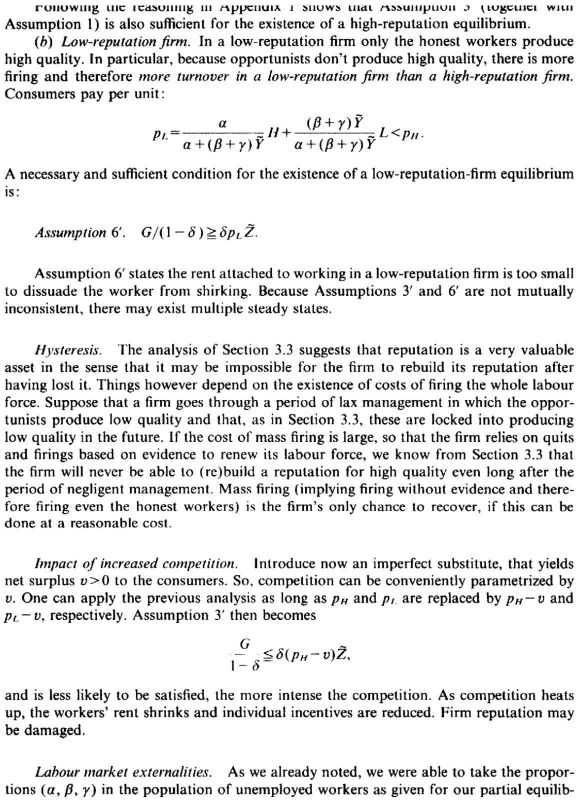
17. In a general equilibrium context, this assumption means that, once fired by a firm, workers are not re-hired by another firm; otherwise dishonest and (depending on the equilibrium) opportunistic workers would be over-represented in the labour market compared to the prior distribution. We could alternatively have

(b) High-reputation firm equilibrium

11.7 Firm hysteresis, competition, and conclusion

Read: The first panel shows the low-reputation firm condition, the hysteresis logic, and the competition result. The second panel gives the conclusion: individual and collective reputations are jointly determined, stereotypes can persist through inherited collective reputation, and anti-corruption policies or amnesties can matter by shifting reputational histories.

Economic message: A firm's bad reputation can be self-confirming, and competition may reduce the rent needed to discipline workers. The same logic that makes corruption persistent also makes quality reputation fragile.



Following the reasoning in Appendix 1 shows that Assumption 5′ (together with Assumption 1) is also sufficient for the existence of a high-reputation equilibrium.

(*b*) *Low-reputation firm.* In a low-reputation firm only the honest workers produce high quality. In particular, because opportunists don't produce high quality, there is more firing and therefore *more turnover in a low-reputation firm than a high-reputation firm.* Consumers pay per unit:

$$p_L = \frac{\alpha}{\alpha + (\beta+\gamma)\tilde{Y}} H + \frac{(\beta+\gamma)\tilde{Y}}{\alpha + (\beta+\gamma)\tilde{Y}} L < p_H.$$

A necessary and sufficient condition for the existence of a low-reputation-firm equilibrium is:

Assumption 6′. $G/(1-\delta) \geqq \delta p_L \tilde{Z}$.

Assumption 6′ states the rent attached to working in a low-reputation firm is too small to dissuade the worker from shirking. Because Assumptions 3′ and 6′ are not mutually inconsistent, there may exist multiple steady states.

Hysteresis. The analysis of Section 3.3 suggests that reputation is a very valuable asset in the sense that it may be impossible for the firm to rebuild its reputation after having lost it. Things however depend on the existence of costs of firing the whole labour force. Suppose that a firm goes through a period of lax management in which the opportunists produce low quality and that, as in Section 3.3, these are locked into producing low quality in the future. If the cost of mass firing is large, so that the firm relies on quits and firings based on evidence to renew its labour force, we know from Section 3.3 that the firm will never be able to (re)build a reputation for high quality even long after the period of negligent management. Mass firing (implying firing without evidence and therefore firing even the honest workers) is the firm's only chance to recover, if this can be done at a reasonable cost.

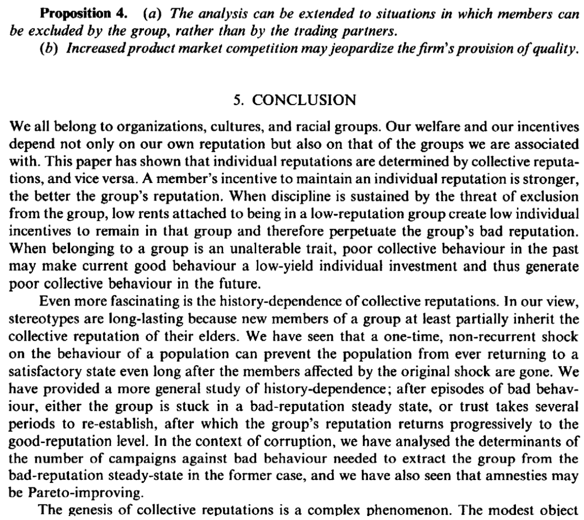
Impact of increased competition. Introduce now an imperfect substitute, that yields net surplus $v>0$ to the consumers. So, competition can be conveniently parametrized by v. One can apply the previous analysis as long as p_H and p_L are replaced by $p_H - v$ and $p_L - v$, respectively. Assumption 3′ then becomes

$$\frac{G}{1-\delta} \leqq \delta(p_H - v)\tilde{Z},$$

and is less likely to be satisfied, the more intense the competition. As competition heats up, the workers' rent shrinks and individual incentives are reduced. Firm reputation may be damaged.

Labour market externalities. As we already noted, we were able to take the proportions (α, β, γ) in the population of unemployed workers as given for our partial equilib-

(a) Firm hysteresis and competition



Proposition 4. (*a*) ***The analysis can be extended to situations in which members can be excluded by the group, rather than by the trading partners.***

(*b*) ***Increased product market competition may jeopardize the firm's provision of quality.***

5. CONCLUSION

We all belong to organizations, cultures, and racial groups. Our welfare and our incentives depend not only on our own reputation but also on that of the groups we are associated with. This paper has shown that individual reputations are determined by collective reputations, and vice versa. A member's incentive to maintain an individual reputation is stronger, the better the group's reputation. When discipline is sustained by the threat of exclusion from the group, low rents attached to being in a low-reputation group create low individual incentives to remain in that group and therefore perpetuate the group's bad reputation. When belonging to a group is an unalterable trait, poor collective behaviour in the past may make current good behaviour a low-yield individual investment and thus generate poor collective behaviour in the future.

Even more fascinating is the history-dependence of collective reputations. In our view, stereotypes are long-lasting because new members of a group at least partially inherit the collective reputation of their elders. We have seen that a one-time, non-recurrent shock on the behaviour of a population can prevent the population from ever returning to a satisfactory state even long after the members affected by the original shock are gone. We have provided a more general study of history-dependence; after episodes of bad behaviour, either the group is stuck in a bad-reputation steady state, or trust takes several periods to re-establish, after which the group's reputation returns progressively to the good-reputation level. In the context of corruption, we have analysed the determinants of the number of campaigns against bad behaviour needed to extract the group from the bad-reputation steady-state in the former case, and we have also seen that amnesties may be Pareto-improving.

The genesis of collective reputations is a complex phenomenon. The modest object

(b) Conclusion and takeaway

12 Short summary

- The paper develops a formal theory of collective reputation as the aggregation of individual reputations.
- Individual past behavior is observed with noise, so group history helps outsiders infer individual behavior.
- A clean individual record can be insufficient when the group has a bad reputation.
- The direct-exclusion model shows how corruption can persist because new members inherit distrust created by old members.
- The model can have both low- and high-corruption steady states.
- A one-time corruption shock can permanently move the economy to high corruption if suspicion lasts long enough and waiting for trust is too costly.
- Short anti-corruption campaigns may fail because they do not rebuild trust; longer campaigns can work by allowing suspicion to fade.
- Amnesty is harmful in steady state but can be Pareto-improving after an out-of-steady-state reputational shock.
- The firm-quality application shows that firm reputation can discipline workers through internal exclusion.

- Competition can weaken firm reputation by reducing worker rents and therefore reducing incentives to maintain quality.

13 Conclusion

13.1 Core conclusion

Tirole shows that individual and collective reputations are jointly determined. A group's reputation affects each member's incentives, while members' behavior feeds back into the group's reputation. This feedback can create strong persistence: bad group behavior can continue long after the original bad actors have exited.

13.2 Economic intuition

The key intuition is that reputation is an investment whose return depends on trust. If outsiders already distrust the group, individual good behavior has a low private return because even a clean record may not restore access to the efficient task. This makes bad behavior rational, which validates outsiders' distrust. Conversely, when the group has a good reputation, individual reputation is valuable, and opportunists have incentives to behave well.

13.3 Incremental contribution revisited

The paper's contribution is not simply to show that beliefs can be persistent. It explains why beliefs and incentives coevolve. Stereotypes are not imposed as exogenous prejudices or common traits; they arise because past collective behavior changes the payoff from current individual behavior.

13.4 Policy implication

Policy must change both behavior and beliefs. A one-period crackdown can stop corruption temporarily but may not change the reputational state. A sustained campaign works by allowing enough time for clean behavior to become credible. Amnesty can work after a shock because it removes inherited stigma, though it is harmful if used in a well-functioning low-corruption steady state.

13.5 Limitations

The paper is a parsimonious theory model. It abstracts from many institutional details: endogenous monitoring, political incentives, heterogeneous principals, network structures, formal legal punishment, and the internal hierarchy of real firms. It is strongest as a conceptual framework for reputational persistence, not as a quantitative model of any particular country or firm.

13.6 Takeaway

Collective reputation is a form of social capital. Once damaged, it can be difficult to rebuild because new members inherit the beliefs created by old behavior. This provides a powerful explanation for persistent corruption, persistent stereotypes, and fragile firm-quality reputations.